EMPAGLIFLOZIN

Euglycaemic DKA

Heart & kidney protection

CHEMICAL NAME (2S,3R,4R,5S,6R)-2-[4-chloro-3-({4-[(3S)-oxolan-3-yloxy]phenyl}methyl)phenyl]-6-(hydroxymethyl)oxane-3,4,5-triol

CLASS Sodium–glucose co-transporter 2 (SGLT2) inhibitor (gliflozin)

TRADE NAMES Jardiance · Synjardy (with metformin) · Glyxambi (with linagliptin)

THERAPEUTIC USES / INDICATIONS

- Type 2 diabetes (especially with cardiovascular disease, heart failure or CKD).
- Heart failure (reduced and preserved EF); with or without diabetes.
- Chronic kidney disease (slows progression).

ADMINISTRATION

- **PO** once daily in the morning, with or without food.
- Stop 3 days before major surgery or during acute illness, dehydration, ketosis (sick-day rules).

DOSAGES

- **Diabetes:** 10 mg once daily, may increase to 25 mg.
- **Heart failure / CKD:** 10 mg once daily.
- Glycaemic efficacy ↓ when eGFR <45 (but cardio-renal benefit persists to eGFR ≥20).

PHARMACOKINETICS

- **Onset:** 1.5 h peak. **Duration:** 24 h.
- Metabolised by glucuronidation.
- Excreted in urine and faeces. **t½:** 12 h.

PHARMACODYNAMICS

- Blocks **SGLT2** in proximal tubule → ↓ renal glucose reabsorption → glycosuria (~70 g/day), osmotic diuresis and natriuresis.
- ↓ BP, weight; cardio-renal protection; low hypoglycaemia risk.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Genital mycotic infections (thrush, balanitis).
- Polyuria, thirst.
- UTI.

ADVERSE EFFECTS (SERIOUS)

- Euglycaemic diabetic ketoacidosis (ketosis with near-normal glucose); surgery, fasting, illness, alcohol, low-carb diets.
- Volume depletion, hypotension, AKI (elderly, diuretics).
- Fournier's gangrene (necrotising perineal fasciitis) — rare.
- Lower-limb amputation concern (canagliflozin class signal).

CONTRAINDICATIONS & CAUTIONS

- Type 1 diabetes (DKA risk); DKA.
- Hypersensitivity.
- eGFR <20 (initiation) or dialysis.
- **Caution:** volume depletion, elderly on diuretics, recurrent genital infections.

KEY INTERACTIONS

- **Loop/thiazide diuretics:** ↑ dehydration, hypotension.
- **Insulin, sulfonylureas:** hypoglycaemia (reduce dose).
- **Lithium:** ↓ levels.

NURSING MANAGEMENT

- Check eGFR, volume status, BP.
- Check ketones (blood) if unwell, even if glucose normal.
- Withhold perioperatively and in acute illness per protocol.
- Genital hygiene advice; monitor for UTI.
- Monitor weight and fluid status in heart failure.

PATIENT EDUCATION

- Keep genital area clean and dry; report itching or discharge.
- Stop during vomiting, diarrhoea, fasting or before surgery (sick-day rules).
- Seek urgent help for nausea, vomiting, abdominal pain, breathlessness (DKA) — even if glucose normal.
- Report pain, swelling or redness in genital area with fever.

KEY POINTS

- SGLT2 inhibitor; glucose in urine.
- Euglycaemic DKA — check ketones; sick-day rules.
- 10 mg OD; HF and CKD benefit.
- Genital thrush; Fournier's gangrene.

SOURCES CHECKED **[614]** Empagliflozin 10 mg film-coated tablets (previously known as Jardiance) (UK SmPC) **[615]** Jardiance, oral (US label) **[616]** Jardiance, oral (US label, Redpharm Drug)

Full references p. 432–457